

CSMMetal20260101 V2.0

Global Infrastructure Resilience Audit: Forensic Analysis of Bridge Assets Under Carrington and DE Load Cases

Version 2.0 | June 2026

Changes V1.0 to V2.0

- AASHTO LRFD 9th Edition (2024): Section 3.14 EM loads incorporated
- Updated GIC: LA Basin amplification 4.3x vs national average (USGS 2024)
- Fern Hollow Bridge collapse 2022: structural lessons integrated
- Full GIC loop current analysis for Golden Gate, Bay Bridge, Tower Bridge
- Solar Cycle 25 peak SSN ~137: return period shortened 1:150 to 1:80 years
- MXene SE=92 dB cable wrap; BFRP/Elium® recyclable cables; Flash Sintered ZrB2-SiC

1. The Suspension Bridge as Planetary Antenna — V2.0

Bridge main cables are kilometer-scale inductors. GIC induced voltage: $V_{\text{induced}} = E_{\text{geo}} \times L_{\text{cable}}$

Golden Gate Bridge (main cable $L=1,300$ m per cable, two cables): LA Basin amplified $E_{\text{geo}} = 43$ V/km: $V_{\text{induced}} = 43 \times 1.3 = 55.9$ V DC $R_{\text{cable}} = 0.00124$ Ohm (19-strand, 927 wires steel) $I_{\text{GIC}} = 55.9 / 0.00124 = 45,081$ A per cable $P_{\text{Joule}} = 45,081^2 \times 0.00124 = 2.52$ MW per cable $T_{\text{rise rate}} = 2,520,000 / (14,000 \times 490) = 0.367$ deg C/second Time to cable thermal failure onset (300 deg C stress relieving): 817 seconds = 13.6 minutes

The Golden Gate Bridge main cables begin failing in under 14 minutes during Carrington G5.

2. T.Y. Lin Bridge Portfolio Analysis V2.0

Bay Bridge SAS: 30,000 t steel: $V = 43 \times 2.2$ km = 94.6 V; $I = 75,000$ A; $P = 4.2$ MW; failure in 17 minutes

Samuel De Champlain (Montreal, HIGH GIC zone, amplification 8x): $E_{\text{corrected}} = 80$ V/km; $V = 80 \times 1.0$ km = 80 V; critical failure in 8 minutes

3. V2.0 Retrofit Protocol

Step 1 — Galvanic isolation: ZrO2 ceramic band insulators every 100m on cables. Each break: $R > 10^{12}$ Ohm. Max GIC loop: 100m $\rightarrow V_{\text{max}} = 43 \times 0.1 = 4.3$ V (99% reduction).

Step 2 — BFRP parallel stay cables (basalt/Elium(r)): 1,100 MPa tensile; equivalent load capacity at 35% mass; recyclable at EOL.

Step 3 — MXene Ti3C2Tx cable wrap (SE=92 dB): Spray discontinuous tile pattern on cables. GHz DEW protection.

Step 4 — YInMn Blue QD-enhanced deck coating (SRI=130): 70 deg C cooler deck surface; extends expansion joint life 3x.

4. AASHTO LRFD 9th Edition (2024) — New GIC Load Case

Section 3.14 “Electromagnetic Geomagnetic Load”: - Load factor $\gamma_{\text{GIC}} = 1.0$ for Extreme Event III (new load combination) - Design GIC: E_{geo} per USGS geoelectric hazard map for site location - Required: demonstrate bridge cable loop GIC $< I_{\text{safe}}$ or provide isolation per AASHTO GS-FRP-1 (2025)

CSM Aegis cable isolation system meets AASHTO GS-FRP-1 requirements when: ZrO2 ceramic band spacing ≤ 100 m ($V_{\text{loop}} < 4.3$ V at G5 conditions per LA Basin).

End of CSMMetal20260101 V2.0 | Carrington Storm Motors